

Juhyun Lee, Ph.D.

Department of Electrical and Computer Engineering, University of California San Diego

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Research Interests

Soft implantable and wearable devices

Neural interfaces for sensing and stimulation

Wireless bioelectronic systems

Education

KAIST

Daejeon, Korea

Ph.D. in Electrical Engineering (GPA: 4.08/4.3)

Mar. 2019 – Aug. 2025

Thesis: Development of wireless optoelectronic brain implants and systems
for broad applications in optogenetics

KAIST

Daejeon, Korea

B.S. in Electrical Engineering (Cum Laude, GPA: 3.79/4.3)

Mar. 2015 – Feb. 2019

Work Experience

Postdoctoral Scholar

La Jolla, CA, USA

Department of Electrical and Computer Engineering, UCSD

Aug. 2026 – present

Advisor: Prof. Shadi Dayeh

Postdoctoral Researcher

Daejeon, Korea

School of Electrical Engineering, KAIST

Sep. 2025 – Jul. 2026

Advisor: Prof. Jae-Woong Jeong

Visiting Student Researcher

Stanford, CA, USA

Department of Chemical Engineering, Stanford University

Sep. 2022 – Feb. 2023

Advisor: Prof. Zhenan Bao

Honors and Awards

InnoCORE Postdoctoral Fellowship, KAIST

May. 2026 – Jul. 2026

BK21 Postdoctoral Fellowship, KAIST

Sep. 2025 – Apr. 2026

Graduate Student Outstanding Paper Award, KAIST

Nov. 2024

Government-Sponsored Scholarship, KAIST

Mar. 2019 – Aug. 2025

BK21 Financial Support for Overseas Long-Term Training, KAIST

Sep. 2022 – Feb. 2023

Outstanding Teaching Assistant Award, KAIST

Apr. 2021

Publications

[†] These authors contributed equally to the work.

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- [1] C. Y. Kim[†], **J. Lee[†]**, E. Y. Jeong, Y. Jang, H. Kim, B. Choi, D. Han, Y. Oh, J.-W. Jeong (2025). “Wireless technologies for wearable electronics: A review”. *Advanced Electronic Materials*, 11(10), 2400884 [Featured on the Front Cover]
- [2] H. Kim[†], **J. Lee[†]**, U. Heo, D. Jayashankar, K.-C. Agno, Y. Kim, C. Y. Kim, Y. Oh, S.-H. Byun, B. Choi, H. Jeong, W.-H. Yeo, Z. Li, S. Park, J. Xiao, J. Kim, J.-W. Jeong (2024). “Skin preparation-free, stretchable microneedle adhesive patches for reliable electrophysiological sensing and exoskeleton robot control”. *Science Advances*, 10(3), eadk5260
- [3] K. E. Parker[†], **J. Lee[†]**, J. R. Kim[†], C. Kawakami, C. Y. Kim, R. Qazi, K.-I. Jang, J.-W. Jeong, J. G. McCall (2023). “Customizable, wireless, and implantable optogenetic probe design and fabrication via 3D printing”. *Nature Protocols*, 18(1), 3-21 [Featured on the Cover]
- [4] **J. Lee[†]**, K. E. Parker[†], C. Kawakami[†], J. R. Kim[†], R. Qazi, J. Yea, S. Zhang, C. Y. Kim, J. Bilbily, J. Xiao, K.-I. Jang, J. G. McCall, J.-W. Jeong (2020). “Rapidly customizable, scalable 3D-printed wireless optogenetic probes for versatile applications in neuroscience”. *Advanced Functional Materials*, 30(46), 2004285 [Featured on the Back Cover]
- [5] H. Kim[†], S. K. Bang[†], S.-Y. Kang, S. W. Kim, J.-S. Yang, D. Kim, C. Y. Kim, S. Han, Y. Jang, **J. Lee**, K.-C. Agno, J.-B. Yoon, S. Lee, J.-W. Jeong (2026). “Wireless IoT-enabled microneedle electroceutical for personalized and connected pain management”. *Nature Communications*, accepted
- [6] C. Y. Kim[†], K. E. Parker[†], E. Y. Jeong, D. Han, Y. Jang, S.-Y. Joo, **J. Lee**, C. Chiangchaisakunthai, R. Qazi, J. Chung, S. Ha, J.-W. Jeong, J. G. McCall (2026). “Protocol to design and implement scalable wireless network system for high-throughput, automated behavioral and circuit neuroscience in mice”. *STAR Protocols*, 7(2), 104453
- [7] I. Kang[†], J. Bilbily[†], C. Y. Kim, C. Shi, M. K. Madasu, E. Y. Jeong, K. E. Parker, D. A. Kwon, B.-J. Jung, J.-S. Yang, **J. Lee**, N. D. L. Kabbaj, W. Lee, J.-B. Yoon, R. Al-Hasani, J. Xiao, J. G. McCall, J.-W. Jeong (2025). “Wireless modular implantable neural device with one-touch magnetic assembly for versatile neuromodulation”. *Advanced Science*, 12(4), 2406576
- [8] S. Oh[†], S. Lee[†], S. W. Kim, C. Y. Kim, E. Y. Jeong, **J. Lee**, D. A. Kwon, J.-W. Jeong (2024). “Softening implantable bioelectronics: material designs, applications, and future directions”. *Biosensors and Bioelectronics*, 116328
- [9] G.-H. Lee[†], H. Kim[†], **J. Lee**, J.-Y. Bae, M. Kim, C. Yang, S. Park, H. Kang, S.-K. Kang, J. Kang, Z. Bao, J.-W. Jeong, S. Park (2023). “Large-area photo-patterning of initially conductive EGaIn particle-assembled film for soft electronics”. *Materials Today*, 67, 84-94
- [10] S.-H. Byun[†], J. Y. Sim[†], Z. Zhou, **J. Lee**, R. Qazi, M. C. Walicki, K. E. Parker, M. P. Haney, S. H. Choi, A. Shon, G. B. Gereau, J. Bilbily, S. Li, Y. Liu, W.-H. Yeo, J. G. McCall, J. Xiao, J.-W. Jeong (2019). “Mechanically transformative electronics, sensors, and implantable devices”. *Science advances*, 5(11), eaay0418

Conference Presentations

- [1] **I. Lee**, K. E. Parker, J. G. McCall, J.-W. Jeong (2026). "Wireless, spatially-matched optoelectronic neural interfaces for simultaneous optogenetics and electrophysiology". *The 28th Korean MEMS Conference*, invited
- [2] **I. Lee**, K. E. Parker, J. G. McCall, J.-W. Jeong (2025). "Spatially-matched optoelectronic neural probes for simultaneous optogenetics and electrophysiology". *2025 Fall Conference of the Korean Sensors Society*, poster presentation
- [3] **I. Lee**, K. E. Parker, J. G. McCall, J.-W. Jeong (2024). "Wireless optoelectronic neural interfaces for simultaneous and spatially-matching optogenetics and electrophysiology". *The 12th World Biomaterials Congress*, poster presentation
- [4] **I. Lee**, K. E. Parker, C. Kawakami, J. R. Kim, R. Qazi, J. Yea, S. Zhang, C. Y. Kim, J. Bilbily, J. Xiao, K.-I. Jang, J. G. McCall, J.-W. Jeong (2021). "Customizable and rapidly manufacturable 3D-printed neural probes for wireless optogenetics". *2021 Virtual MRS Spring Meeting & Exhibit*, oral presentation
- [5] H. Kim, **I. Lee**, U. Heo, D. Jayashankar, K.-C. Agno, Y. Kim, C. Y. Kim, Y. Oh, S.-H. Byun, B. Choi, H. Jeong, W.-H. Yeo, Z. Li, S. Park, J. Xiao, J. Kim, J.-W. Jeong (2024). "Skin-preparation-free, stretchable microneedle adhesive patches for high-fidelity electrophysiological sensing and exoskeleton robot control". *2024 MRS Spring Meeting & Exhibit*, oral presentation
- [6] H. Kim, **I. Lee**, J.-W. Jeong (2022). "Adhesive, air-permeable, stretchable conductive sensors for long-term electrophysiological signal monitoring". *The 6th International Conference on Active Materials and Soft Mechatronics*, poster presentation
- [7] S.-H. Byun, J. Y. Sim, Z. Zhou, **I. Lee**, R. Qazi, M. C. Walicki, K. E. Parker, M. P. Haney, S. H. Choi, A. Shon, G. B. Gereau, J. Bilbily, S. Li, Y. Liu, W.-H. Yeo, J. G. McCall, J. Xiao, J.-W. Jeong (2020). "Mechanically transformative electronics enabled by Gallium-elastomer composite". *2020 BMES Annual Meeting*

Teaching Experience

Research Mentorship, KAIST

Mentored 6 undergraduate and 3 graduate students Jun. 2019 – present

Teaching Assistant, KAIST

EE405 Electronics Design Lab (Device) Spring 2020, Fall 2021, Spring 2022, Fall 2023

EE566 MEMS in EE Perspective Spring 2021

EE405 Electronics Design Lab (Circuit) Fall 2019

Freshman Tutoring Program, KAIST

General Physics I Spring 2016, Fall 2017, Spring 2018

Extracurricular Experience

Futsal Club, KAIST

Sep. 2017 – Feb. 2019

Classical Guitar Club, KAIST

Mar. 2015 – Feb. 2019

Language Exchange Program (German), KAIST

Sep. 2017 – Dec. 2017

Exchange student, Karlsruhe Institute of Technology

Mar. 2017 – Aug. 2017